

Machine Learning (CSP 5211)

Course Overview

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Machine Learning Overview

*Welcome to the world
of
Machine Learning*

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Machine Learning Overview

- Introduction to Machine Learning (ML)
- Prerequisites
- Course Objectives
- Course Content
- Learning Outcomes
- Text Books
- Grading Rubric
- Course Web Page
- Office Hours

What is Machine Learning?

From data to intelligence

Arthur Samuel's Definition (1959)

“Field of study that gives computers the ability to learn without being explicitly programmed.”

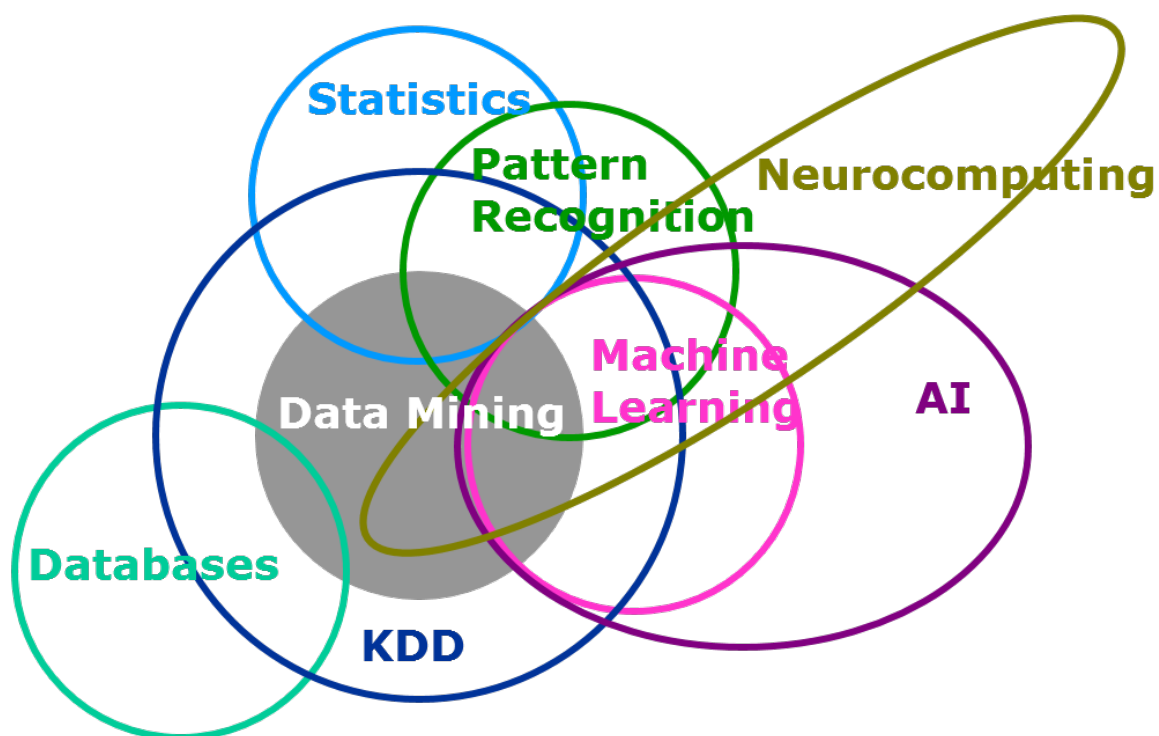
Key Components:

- **Data:** The raw material for learning
- **Model:** Mathematical representation of patterns
- **Learning Algorithm:** Method to extract patterns
- **Evaluation:** Measuring performance

“A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .”

–Tom Mitchell, Carnegie Mellon University

So if you want your program to predict, for example, traffic patterns at a busy intersection (task T), you can run it through a machine learning algorithm with data about past traffic patterns (experience E), if it has successfully “learned”, it will then do better at predicting future traffic patterns (performance measure P).



Why Machine Learning Now?

Perfect storm

Driving Forces:

- **Data Explosion:** 2.5 quintillion bytes daily
- **Compute Power:** GPU/TPU advancements
- **Algorithmic Innovation:** Deep learning breakthroughs
- **Storage Economics:** Cloud computing
- **Applications:** Healthcare, finance, autonomous systems

Market Growth:

Year	Market Size (USD)
2018	7.3B
2020	15.4B
2022	31.4B
2024	67.0B

[Table:](#) Machine Learning Market Growth

Types of Machine Learning

Learning paradigms

Three Main Categories

- 1 **Supervised Learning:** Learn from labeled examples
- 2 **Unsupervised Learning:** Discover patterns in unlabeled data
- 3 **Reinforcement Learning:** Learn through trial and error

Course Focus

This course covers **supervised** and **unsupervised** learning in depth.

Emerging Paradigm

We'll also touch on **self-supervised learning** used in foundation models like GPT and DALL-E.

Mathematical Foundations

What you need to know

Linear Algebra

- Vectors
- Matrices
- Eigenvalues
- SVD

Probability & Statistics

- Distributions
- Bayes' Theorem
- Expectation
- Maximum Likelihood

Calculus

- Derivatives
- Gradients
- Optimization
- Chain Rule

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Programming Skills

Technical requirements

Required Proficiency

- **Python:** Primary language for course
- **Numpy/Scipy:** Numerical computing
- **Matplotlib:** Data visualization
- **Jupyter:** Interactive development
- **Basic Algorithms:** Data structures and complexity

Recommended Resources

- [Python.org](https://python.org) - Official documentation
- [NumPy.org](https://numpy.org) - Array operations
- [Matplotlib.org](https://matplotlib.org) - Visualization

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What Will You Learn?

Course goals and aspirations

Core Objectives

- 1 Understand fundamental ML algorithms and their mathematical foundations
- 2 Implement algorithms from scratch and using libraries
- 3 Evaluate model performance using appropriate metrics
- 4 Apply ML to real-world problems with ethical considerations
- 5 Prepare for advanced study or industry applications

Beyond the Syllabus

- **Critical Thinking:** When to use which algorithm
- **Troubleshooting:** Debugging models that don't work
- **Communication:** Explaining ML concepts to non-experts

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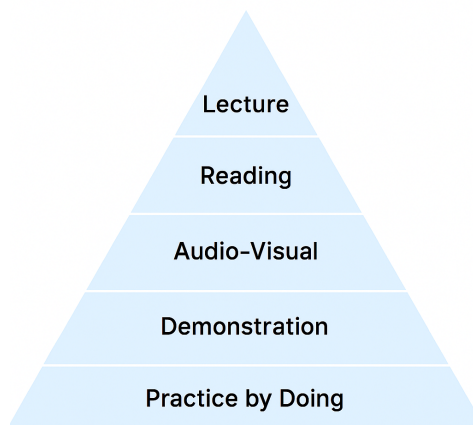
Course Philosophy

How we approach learning

Three Pillars:

- 1 **Theory:** Mathematical foundations
- 2 **Implementation:** Hands-on coding
- 3 **Application:** Real-world problems

Learning Pyramid



Bloom's Taxonomy

We aim for **analysis**, **application**, **evaluation**, and **creation**-not just remembering facts.

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Course Structure

Four main modules

Module 1

Supervised Learning

- Linear regression
- Classification
- Logistic regression
- GLMs
- Kernel methods
- SVM

Module 2

Neural Networks

- Non-linear models
- Modern architectures
- Backpropagation
- Vectorization

Module 3

Generalization

- Overfitting
- Regularization
- Model selection
- Evaluation

Module 4

Unsupervised Learning

- Clustering
- EM algorithm
- PCA
- ICA
- Self-supervision

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Detailed Syllabus

What each module covers

Module 1: Supervised Learning (Weeks 1-4)

- **Week 1:** Linear regression, loss functions, gradient descent
- **Week 2:** Classification, logistic regression, decision boundaries
- **Week 3:** Generalized linear models, exponential family
- **Week 4:** Kernel methods, support vector machines

Module 2: Neural Networks (Weeks 5-8)

- **Week 5:** From linear to non-linear models, activation functions
- **Week 6:** Modern NN architectures (CNNs, RNNs, Transformers)
- **Week 7:** Backpropagation, computational graphs
- **Week 8:** Optimization, vectorization, GPU acceleration

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Detailed Syllabus (Continued)

What each module covers

Module 3: Generalization (Weeks 9-10)

- **Week 9:** Bias-variance tradeoff, overfitting, underfitting
- **Week 10:** Regularization techniques (L1, L2, dropout)
- **Week 10:** Cross-validation, hyperparameter tuning

Module 4: Unsupervised Learning (Weeks 11-14)

- **Week 11:** Clustering, k-means, hierarchical clustering
- **Week 12:** Expectation-Maximization algorithm
- **Week 13:** Dimensionality reduction (PCA, ICA)
- **Week 14:** Self-supervised learning, foundation models

Knowledge and Skills

What you'll be able to do

Technical Skills:

- Implement ML algorithms in Python
- Preprocess and visualize data
- Train, validate, and test models
- Use libraries like scikit-learn, TensorFlow/PyTorch
- Deploy simple ML models

Analytical Skills:

- Select appropriate algorithms for problems
- Diagnose model performance issues
- Interpret model outputs
- Assess ethical implications
- Read and understand ML research papers

Direct Applications:

- Machine Learning Engineer
- Data Scientist
- AI Researcher
- Business Intelligence Analyst
- ML Operations (MLOps)

Further Education

- Advanced ML courses (Deep Learning, RL, NLP)
- Graduate studies (Masters/PhD)
- Specialized certifications

Success Tips

How to excel in this course

Do:

- Attend lectures regularly
- Start assignments early
- Form study groups
- Practice coding daily
- Ask questions
- Explore beyond syllabus

Avoid:

- Cramming before exams
- Copying code without understanding
- Skipping mathematical derivations
- Ignoring data visualization
- Underestimating project time

Textbooks

- ① Class notes will be provided (Source:CS229)
- ② Theodoridis, Sergios. Machine Learning: From the Classics to Deep Networks, Transformers, and Diffusion Models. Netherlands, Elsevier Science, 2025.
- ③ Murphy, Kevin P. Machine Learning, Second Edition: A Probabilistic Perspective. United States, MIT Press, 2020.
- ④ Christopher M. Bishop. Pattern Recognition and Machine learning. Springer, 2007.
- ⑤ IEEE Transactions on Pattern Analysis and Machine Intelligence

Assessment

- Assignments (10%)
- Project (10%)
- Midterm (30%)
- Final Exam (50%)

Course Logistics:Communication

- This course will have a web page on [MS Teams](#)
- Use team code:[waimh5k](#) to join the team
- MS Teams will act as an off class interface between you and me
- All the assignments and course study material (if any) will be available on the course web page
- Use MS Teams app for getting quick notification (if any)
- Discussion forum for Q&A
- Email for private questions

Course Logistics: Student Office Hours

- Instructor: Sanjay Singh
- Time: Mondays, and Saturdays, 3–4 PM
- Location: Faculty Room #11, Chamber # 99
School of Computer Engineering, First Floor, AB5
- Email: sanjay.singh@manipal.edu

Course Logistics: Mobile Phone Usage Policy

- Mobile phones must be kept on silent mode during class hours
- Use of phones for texting, calling, or browsing is strictly prohibited during lectures
- Phones may be used only when explicitly permitted for academic activities
- Repeated violations may result in disciplinary action
- Respect the learning environment and minimize distractions for yourself and others



That's It! Enjoy the Course!
Thank You